

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	VineethaKuppala/Rising-Waters
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask

Key Features Implemented	• Flood prediction using a trained XGBoost machine learning model. • User-friendly web interface for entering environmental parameters. • Data preprocessing and feature scaling before prediction. • Real-time prediction using Flask. • Model serialization using Pickle. • Responsive frontend built with HTML and CSS.
Pending / Incomplete Features	• Integration with real-time weather APIs. • Cloud deployment (Render/AWS/Heroku). • User authentication and prediction history. • Interactive analytics dashboard. • Mobile application support.
Setup / Run Instructions	1. Clone or download the repository. 2. Open the project in Visual Studio Code. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Run the application using <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000/</code> in a web browser. 6. Enter the required parameters and click Predict to view the result.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

6	Code is committed to a version control repository	Yes
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Additional Notes / Comments:

The project demonstrates the successful integration of Machine Learning with a Flask-based web application for flood prediction. The code follows a modular structure, making it easier to maintain and extend. Future enhancements include real-time weather data integration, cloud deployment, improved prediction accuracy using larger datasets, and advanced visualization features.